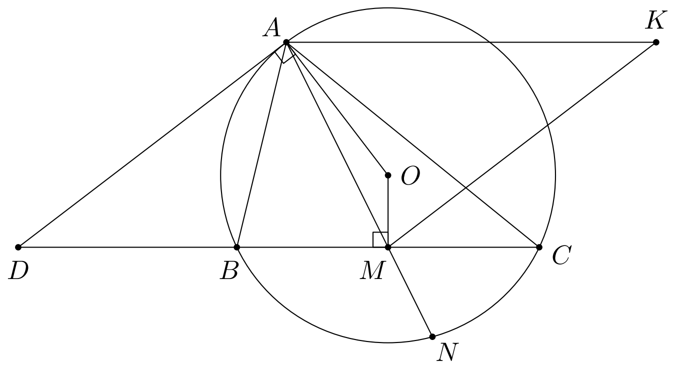
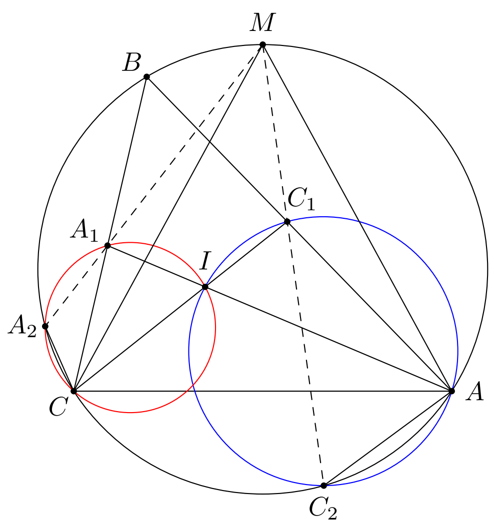
Problem 37 (International Mathematical Excellence Olympiad 2019 Problem 1)**.** Let $\triangle ABC$ be a scalene triangle. The median from A meets BC and the circumcircle of $\triangle ABC$ again at M and N respectively. The tangent at A to the circumcircle of $\triangle ABC$ meets the extension of BC at D. Let K be the point such that $ADMK$ is a parallelogram. Prove that $KA = KN$.



Solution. Let O be the circumcentre of $\triangle ABC$. As $OM \perp DM$, $OM \perp AK$. As $OA \perp AD$, $OA \perp KM$. Therefore, O is the orthocentre of $\triangle AMK$. This implies $OK \perp AM$, and hence OK is the perpendicular bisector of AN since $OA = ON$. Thus, $KA = KN$. $\square$

Problem 38 (Oral Moscow Geometry Olympiad 2014 Grade 10–11 Problem 3)**.** Let A_1 and C_1 be the feet of internal angle bisectors from A and C of $\triangle ABC$ respectively, and let I be the incentre of $\triangle ABC$. The circumcircle of $\triangle AIC_1$ meets the arc AC not containing B of the circumcircle of $\triangle ABC$ at C_2. The circumcircle of $\triangle CIA_1$ meets the arc BC not containing A of the circumcircle of $\triangle ABC$ at A_2. Prove that the lines A_1A_2 and C_1C_2 intersect on the circumcircle of $\triangle ABC$.



Solution. Let M be the midpoint of the arc ABC of the circumcircle of $\triangle ABC$. We claim that A_1A_2 and C_1C_2 intersect at M.

Firstly, $\angle CA_2A_1 = \angle CIA = 90^\circ + \dfrac{B}{2} = 180^\circ - \angle MAC = \angle CA_2M$, so that A_2, A_1, M are collinear.

Secondly, $\angle AC_2C_1 = \angle AIC_1 = 90^\circ - \dfrac{B}{2} = \angle ACM = \angle AC_2M$, so that C_2, C_1, M are collinear.

This completes the proof. $\square$

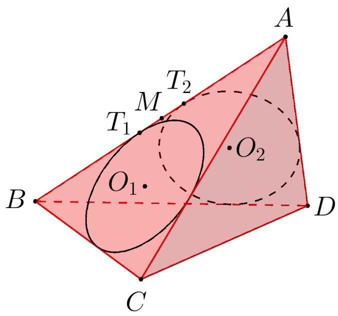
Problem 39 (Oral Moscow Team MO 2017 Grade 11 Problem 3)**.** There is a sphere in the space. Let A, B, C, D be four points in the space such that each of AB, BC, CD, DA is tangent to the sphere. Prove that all tangential points are coplanar.

Solution. Let the sphere touch AB, BC, CD, DA at W, X, Y, Z respectively. Then $AW = AZ$, $BW = BX$, $CX = CY$ and $DY = DZ$. If A, B, C, D are coplanar, the result is obvious. If A, B, C, D are non-coplanar, since

$$\frac{AW}{WB} \times \frac{BX}{XC} \times \frac{CY}{YD} \times \frac{DZ}{ZA} = \frac{AW}{ZA} \times \frac{BX}{WB} \times \frac{CY}{XC} \times \frac{DZ}{YD} = 1,$$

the points W, X, Y, Z are coplanar by Carnot's theorem. $\square$

Problem 40 (Sharygin MO 2015 Correspondence Problem 23)**.** A tetrahedron $ABCD$ is given. The incircles of $\triangle ABC$ and $\triangle ABD$ with centres O_1 and O_2 touch AB at points T_1 and T_2 respectively. Let Π_{AB} be the plane passing through the midpoint of T_1T_2 and is perpendicular to O_1O_2. Define the planes $\Pi_{AC}, \Pi_{BC}, \Pi_{AD}, \Pi_{BD}, \Pi_{CD}$ analogously. Prove that these 6 planes have a common point.



Solution. Let M be the midpoint of T_1T_2. Let S_D be the sphere having the incircle of $\triangle ABC$ as a great circle. Define S_C, S_B, S_A analogously. Since T_1T_2 is tangent to S_D and S_C, and $MT_1 = MT_2$, the powers of M with respect to S_D and S_C are the same. So M lies on the radical plane of S_D and S_C. Therefore, Π_{AB} is the radical plane of S_D and S_C.

Similarly, the 6 planes are the radical planes of two of the spheres S_A, S_B, S_C, S_D. As the centres of these spheres are non-coplanar, the 6 radical planes intersect at the radical centre of these spheres. $\square$

Remarks: The radical plane of two spheres is the locus of points having equal power with respect to the two spheres. It is a plane if and only if the two spheres do not have the same centre. The radical plane of two spheres is perpendicular to the line joining the centres of the spheres.

Given four spheres with non-coplanar centres, their pairwise radical planes meet at a unique point, called the radical centre of these spheres. Indeed, using the notations in this problem, one can show that the radical planes $\Pi_{AB}, \Pi_{BC}, \Pi_{CD}$ must intersect at a point. This point clearly has equal power with respect to S_A, S_B, S_C, S_D, and so it lies on all the 6 radical planes.